

NPTEL

Denoising Diffusion Probabilistic Model

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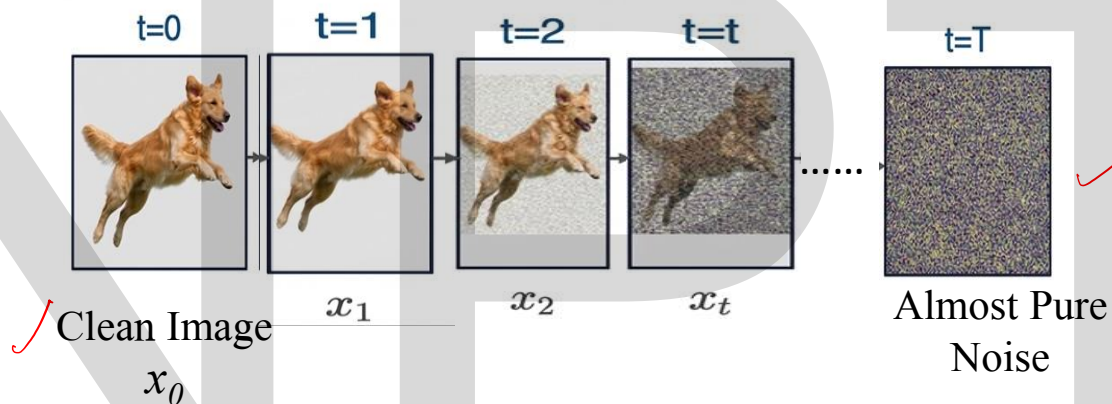
Content:

➤ Forward Diffusion Process

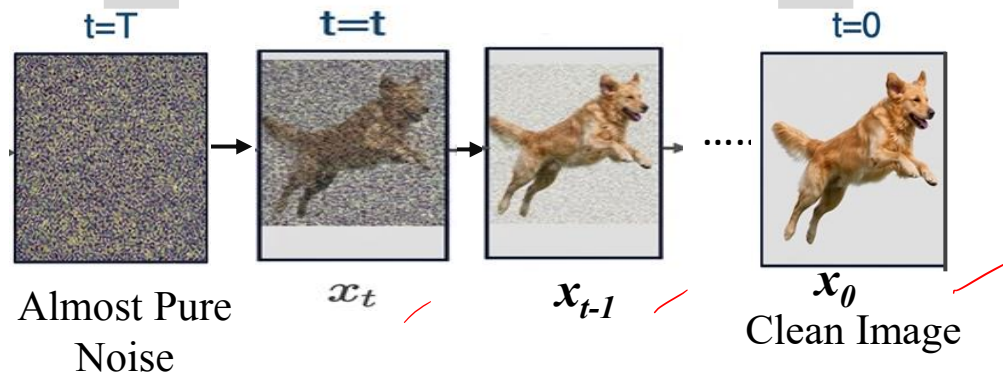
Denoising Diffusion Probabilistic Model(DDPM)

DDPM is a specific formulation of diffusion probabilistic models.

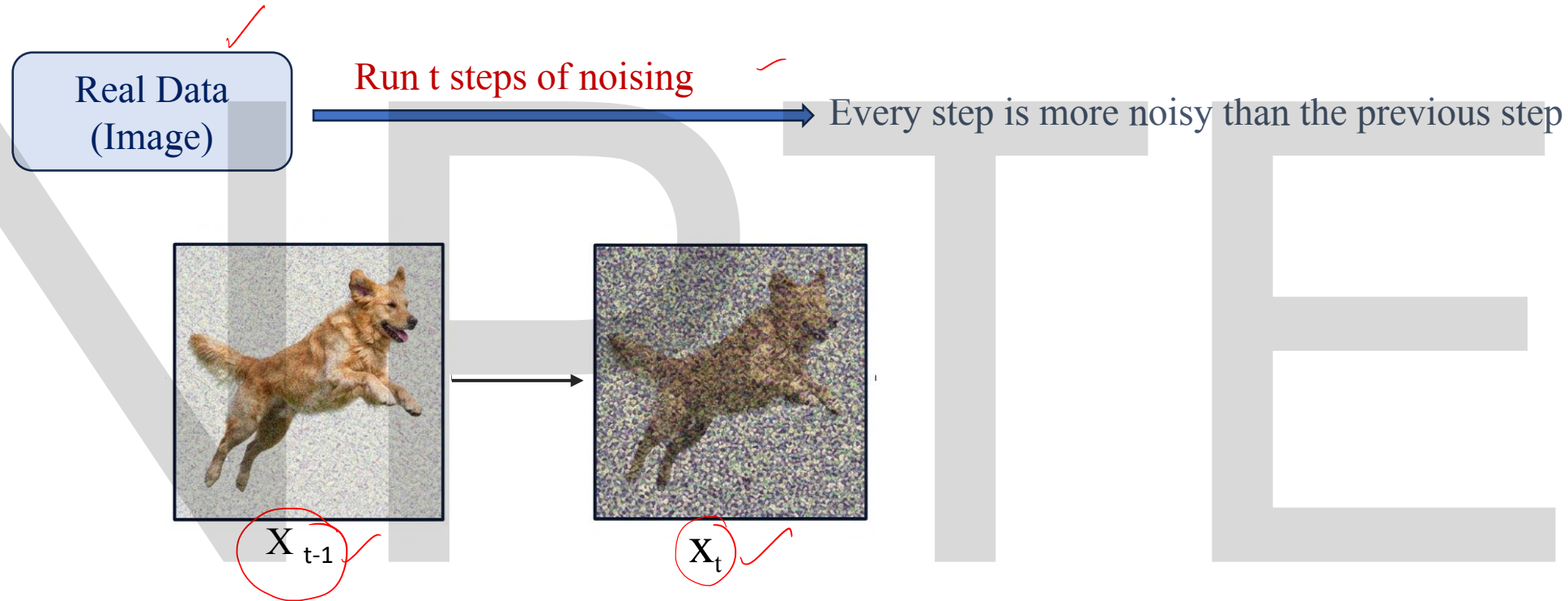
Forward Diffusion Process



Reverse Diffusion Process



Mathematical Formulation of Forward Diffusion Process (Markov Chain)



$q(x_t | x_{t-1})$: Probability of getting x_t given x_{t-1} (Conditional Probability)

$N(x_t; \mu, \Sigma)$ ✓ $\Rightarrow$ x_t is sampled from a Gaussian distribution with mean = μ and covariance = Σ

Forward Diffusion Process

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

Forward diffusion distribution.

Covariance

- ✓ Covariance describes how uncertainty (noise) is distributed across dimensions and how different dimensions are related.

$\beta_t I$: Covariance Term

✓ β_t : Magnitude of noise

I: Identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓

$$\underline{\beta_t I} = \beta_t \underline{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} \beta_t & 0 \\ 0 & \beta_t \end{bmatrix} \quad \checkmark$$

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Covar}(x_1, x_2) \\ \text{Covar}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix} \quad \checkmark$$

Zero Covariance means no relationship between the dimensions

Two variables
 $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Forward Diffusion Process

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

Mean

Covariance

X_t = image at time step t

X_{t-1} = image at time step $(t-1)$

I : Identity Matrix

β_t : controls how much noise is added

- Given the previous image x_{t-1} , the next image x_t is sampled from Gaussian Distribution with mean $\sqrt{1 - \beta_t} x_{t-1}$ and covariance $\beta_t I$.
- The mean is a scaled version of the previous image
- Scaling factor: $\sqrt{1 - \beta_t}$ where β_t is between 0 and 1.

Why scaling the signal with $\sqrt{(1 - \beta_t)}$ is needed?

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

$$\text{Let } \alpha_t = 1 - \beta_t$$

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1 - \alpha_t) I)$$

$$\alpha_t = 1 - \beta_t$$

$$\beta_t = (1 - \alpha_t)$$

A sample can be written as

$$x_t = \underbrace{\sqrt{\alpha_t} x_{t-1}}_{\text{Scaled Signal Strength}} + \underbrace{\sqrt{(1 - \alpha_t)} \varepsilon_t}_{\text{Noise}}$$

Signal is scaled with $\sqrt{\alpha_t}$

With $\varepsilon_t \sim \mathcal{N}(0, 1)$

What happens if the signal is not scaled?

$$\text{If } x_t = x_{t-1} + \sqrt{(1 - \alpha_t)} \varepsilon_t \Rightarrow$$

$$\text{Var}(x_t) = \text{Var}(x_{t-1}) + (1 - \alpha_t)$$

$$\begin{aligned} \text{Var}(x_t) &= \text{Var}(x_{t-1}) + \text{Var}(\sqrt{1 - \alpha_t} \varepsilon_t) \\ &= \text{Var}(x_{t-1}) + \text{Var}(a \varepsilon_t) \\ &= \text{Var}(x_{t-1}) + a^2 \text{Var} \varepsilon_t \\ &= \text{Var}(x_{t-1}) + (1 - \alpha_t) \text{Var} \varepsilon_t \end{aligned}$$

$$a = \sqrt{1 - \alpha_t}$$

- ✓ Variance grows at every step
- ✓ After many steps, the variance would become extremely large and the distribution would "blow up"

Expanding recursively

$$x_1 = \sqrt{\alpha_1} x_0 + \sqrt{(1-\alpha_1)} \epsilon_1$$

$$\begin{aligned} x_2 &= \sqrt{\alpha_2} x_1 + \sqrt{1-\alpha_2} \epsilon_2 \\ &= \sqrt{\alpha_2} \left[\sqrt{\alpha_1} x_0 + \sqrt{(1-\alpha_1)} \epsilon_1 \right] + \sqrt{1-\alpha_2} \epsilon_2 \\ &= \sqrt{\alpha_1 \alpha_2} x_0 + \sqrt{\alpha_2 (1-\alpha_1)} \epsilon_1 + \sqrt{1-\alpha_2} \epsilon_2 \end{aligned}$$

Contains original signal

Gaussian noise

Gaussian noise

$$x_2 = \sqrt{\alpha_1 \alpha_2} x_0 + (\text{sum of Gaussian noise})$$

let's consider $\bar{\alpha}_2 = \alpha_1 \alpha_2$

$$\text{let } Z = \sqrt{\alpha_2 (1-\alpha_1)} \epsilon_1 + \sqrt{1-\alpha_2} \epsilon_2$$

Since ϵ_1 & ϵ_2 are independent Gaussians

$$\begin{aligned} \text{Its Covariance} &= \alpha_2 (1-\alpha_1) I + (1-\alpha_2) I \\ &= (\cancel{\alpha_2} - \alpha_1 \alpha_2 + \cancel{1-\alpha_2}) I = (1 - \alpha_1 \alpha_2) I \\ &= (1 - \bar{\alpha}_2) I \end{aligned}$$

$$x_2 = \bar{\alpha}_2 x_0 + (1 - \bar{\alpha}_2) I$$

Let us define

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

$$\sqrt{\alpha_1 \alpha_2 \alpha_3 \dots \alpha_t} = \sqrt{\prod_{s=1}^t \alpha_s}$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$$

Mean

Variance

Reparameterization Trick

- ✓ Suppose we have $z \sim \mathcal{N}(\mu, \sigma^2)$.
- ✓ Instead of sampling z directly, we write

$$z = \mu + \sigma\epsilon,$$

$$\epsilon \sim \mathcal{N}(0, 1)$$

where

Applying that idea to

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

$$x_t = \underbrace{\sqrt{\bar{\alpha}_t}x_0}_{\text{Remaining Signal Strength}} + \underbrace{\sqrt{(1 - \bar{\alpha}_t)}\epsilon}_{\text{Accumulated noise}}$$

Without reparameterization:

$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_t$
requires simulating t diffusion steps.

With reparameterization:

x_t can be obtained directly in one step for any random timestep t .

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Summary

*Next Session:
Reverse Process in DDPM*